

LuminaX Round 1 corrected evidence addendum

Team: LuminaX

Members: Ahan Sardar and Enakshee Mondal

Date: 10 October 2026

Purpose

This short addendum answers the Round 1 feedback with new analysis. It does not claim that a public result guarantees a private score or selection. Where the new evidence weakens our original submission, we say so and change the conclusion.

What changed

Question from the feedback	Correction	Decision
Were correlated scenario variants treated as independent?	All intervals now resample an entire generated seed, keeping its six variants together and giving each variant equal weight.	The reported history gain remains uncertain.
What do the two directional scores mean?	We define two event probabilities: whether user_a and whether user_b records Yes by the seven-day response deadline. We fit, calibrate and test them on three disjoint seed blocks.	This is a diagnostic estimator, not the matching score.
Are people with sparse profiles served differently?	We report coverage, unserved rate, waits and outcomes in four completeness bands. We also separate arrival timing and observable partner opportunity, then hide 50% and 100% of observed soft fields in controlled paired runs.	A large descriptive service gap exists. The masking tests and service strata test which explanations the evidence can support.
Does guarded cardinality improve completed outcomes?	We isolate the allocator with and without the history signal, then measure MSMI, mutual acceptances, dates, coverage, unserved members and waits.	We retain it only if the completed-outcome evidence supports it.
Is state-order clarification justified?	We compare graph-aware ordering and a stable arrival-day plus opaque-ID order against the old input-order rule. We also permute every observable list and constraint array.	The submitted rule no longer depends on arbitrary JSON member order. Its score effect is reported as uncertain.

Analysis rules

- The primary measure is mutual second-meeting intentions (MSMI) per 100 arrived members. It is a rate, not a probability bounded by 1.
- Each comparison uses identical generated seeds and scenario variants for both methods.
- The six public variants are development, sparse, cold_start, delayed, shift and drift. Each receives equal weight.
- A bootstrap draw samples whole seed groups. It never separates variants that share a generated world.
- New component studies use separate seed blocks. The two masking conditions intentionally reuse the same profile-study worlds so their differences are paired.
- Policy features come only from the observable request. Member IDs, hidden simulator preferences, private seeds and future feedback are never features.
- A 95% interval that includes zero is reported as uncertain.

1. History signal with seed-grouped uncertainty

We reanalysed the saved 20-seed, six-variant comparison. The day-20 history method scored **0.0208 MSMI per 100 higher** than the no-history method across 120 paired episodes. The corrected seed-grouped 95% interval is **-0.0375 to 0.0792**. The interval includes zero.

The variant mean differences were 0.0500 in development, 0.0500 in drift, 0.0250 in shift and 0 in cold-start, delayed and sparse. None was negative in this block, but the overall gain is not established. We therefore retract any wording that treats history as a proven improvement.

Evidence: [results/corrected_history_seed_grouped.json](#)

2. Directional probability definitions and calibration

The old report described directional scores too loosely. The corrected targets are:

- `user_a_recorded_yes_by_response_deadline`: the canonical `user_a` endpoint of an assigned introduction records Yes no later than its seven-day deadline.
- `user_b_recorded_yes_by_response_deadline`: the same event for the canonical `user_b` endpoint.

`user_a` and `user_b` are only canonical pair positions. The same estimator is applied after swapping actor and candidate. A missing response at the deadline is target 0 because the defined Yes-by-deadline event did not happen; it is not interpreted as dislike.

Features are frozen immediately before assignment. Every assignment receives 40 follow-up days, so the holdout contains no right-censored response windows. The model is fit on seeds 6001-6010, calibrated on 6031-6040 and evaluated once on 6101-6110. Each block contains all six variants.

The fresh audit runs the actual submitted `adaptive_greedy` mode. The untouched holdout contains 9,588 directional examples and 3,298 positives. Headline calibration metrics give each scenario variant equal weight; the AUC uses all holdout examples because it is a ranking statistic rather than a scenario mean.

Holdout estimator	Observed Yes	Mean prediction	Brier score	Equal-count ECE	AUC
Logistic model before Platt calibration	34.36%	36.24%	0.22330	0.03391	0.5638
After disjoint Platt calibration	34.36%	36.28%	0.22375	0.04010	0.5638
Constant training-prevalence baseline	34.36%	36.20%	0.22587	0.03578	0.5000

The post-calibration equal-variant Brier score has a seed-grouped 95% interval of **0.21960 to 0.22900**.

Calibration-in-the-large is **+0.0192**, with an interval of **+0.0022 to +0.0328**. The model overpredicts Yes on this holdout. Its paired Brier improvement over the constant baseline is **-0.00212**, with a seed-grouped interval of **-0.00275 to -0.00147**. Platt calibration changes Brier by **+0.00045** versus the raw model, with an interval of **-0.00017 to +0.00105**. The extra calibration step is therefore not supported.

Calibration also differs by profile information available at assignment:

Observed soft fields	Examples	Observed Yes	Mean prediction	Brier	Equal-count ECE
0	2,188	32.91%	34.92%	0.22095	0.04241
1-2	2,300	35.04%	35.47%	0.22696	0.04208
3-6	131	35.88%	35.13%	0.22690	0.14416
7	4,969	34.72%	37.35%	0.22336	0.05357

The 3-6 group is too small for a stable bin estimate. Complete profiles are overpredicted by 2.63 percentage points. Shift and drift also overpredict by 3.16 and 3.00 points. These subgroup checks are diagnostic point estimates, not independent causal comparisons.

The subgroup uncertainty check resamples whole generated worlds. For complete profiles, calibration-in-the-large is **+0.0263**, with a 95% interval from **+0.0113 to +0.0405**. For the small 3-6 group it is **-0.0075**, with a much wider interval from **-0.2021 to +0.1564**. We therefore keep the complete- profile overprediction finding and suppress strong interpretation of the small group.

Calibration also changes with assignment time:

Assignment day	Examples	Observed Yes	Mean prediction	Calibration gap	95% interval for gap	Equal-count ECE
0-19	5,262	36.35%	36.57%	+0.21 points	-1.74 to +2.02 points	2.30%
20-34	2,388	33.17%	36.04%	+2.87 points	-1.21 to +6.26 points	6.20%
35-59	1,938	30.60%	35.95%	+5.36 points	+3.33 to +7.54 points	5.57%

The late-period overprediction is clear in this holdout and is another reason not to put the estimator into the policy. Equal-count ECE is 3.03%, 3.19% and 3.94% with 5, 10 and 20 bins respectively, so the conclusion does not depend on one bin count. We also cross profile completeness, candidate opportunity and assignment time: 12 cells meet the 200-example threshold and 11 are marked `suppressed_small_sample`. No suppressed cell is interpreted.

The fitted model shows weak discrimination and a small, repeatable Brier improvement over the constant baseline. Its calibration-in-the-large remains wrong in the positive direction, subgroup calibration is uneven, and Platt calibration does not improve the raw model. We therefore do not treat this estimator as ready for policy scoring. The result directly supports the feedback that directional calibration still needs development.

These probabilities are not the policy's compatibility score. They are kept outside the runtime unless they beat the constant training-prevalence baseline and remain calibrated on new seed groups.

Evidence: [results/corrected_directional_calibration.json](#)

3. Service by profile completeness

Completeness is the number of the seven soft fields observed when a member first appears. The cohort is measured before masking and never changes later. The unmasked study contains 10 seed groups x 6 variants = 60 episodes. The 12,000 member observations below are correlated across variants, so they are useful descriptive totals, not 12,000 independent worlds.

Observed soft fields	Member observations	Coverage	Unservd	Median wait if served	90th-percentile wait	MSMI member rate
0	3,975	23.80%	76.20%	8.0 days	18.0 days	0.33%
1-2	4,001	23.02%	76.98%	7.0 days	17.0 days	0.42%
3-6	259	26.25%	73.75%	8.0 days	18.0 days	0.39%
7	3,765	50.52%	49.48%	0.0 days	13.0 days	1.35%

Fully complete profiles receive much faster and more frequent service. The 3-6-field band is small, so its outcome rate is especially noisy. Conditioning only on having at least one partner not ruled out at first appearance leaves coverage at 23.83%, 23.03%, 26.25% and 59.99% across the same four bands. This does not remove the gap.

That opportunity measure is deliberately observable and conservative. Missing hard fields leave many partners "not ruled out" even though they may later fail a constraint. The mean count of immediately decidable feasible partners is zero in the first three soft-completeness bands and 0.41 in the complete band. This shows that soft completeness is strongly tied to hard-information readiness in the generator. The saved output therefore includes cells crossing soft completeness, arrival day and observable partner opportunity instead of claiming that soft answers alone caused the service difference.

The new whole-seed subgroup intervals keep that conclusion honest. Coverage is 23.80% (95% interval 22.12%-25.70%) for the zero-field band, 23.02% (19.96%-26.06%) for 1-2 fields, 26.25% (17.91%-37.45%) for the small 3-6 band, and 50.52% (45.00%-55.23%) for complete profiles. The wide 3-6 interval shows why its point estimate cannot carry much weight.

Service is also normalised by actual exposure. Introductions per 100 available- member-days are 1.01, 0.91, 1.09 and 2.42 across the four bands. Observable opportunity exists on 91.02%, 93.20%, 93.28% and 77.95% of available days. When members never served are included through the end of the 60-day decision window, mean days without a first introduction are 45.51, 45.46, 44.78 and 30.32. Repeated service is concentrated too: members with two or more introductions number 534, 462, 37 and 1,062, while the maximum for one member is 8, 7, 5 and 8. These are descriptive service facts, not causal effects.

In the paired 50% masking condition, MSMI changed by **+0.0583** with a seed-grouped 95% interval of **-0.0583 to 0.1750**. Coverage changed by **-0.017 percentage points** with an interval of **-0.167 to 0.117 percentage points**. The coverage gap changed by **-0.19 percentage points**, with an interval of **-0.73 to +0.27 percentage points**. This intervention did not meaningfully reduce the service gap.

When all observed soft fields are hidden from the policy, MSMI changes by **-0.0083**, with a seed-grouped 95% interval of **-0.1500 to 0.1250**. Coverage changes by **-0.117 percentage points**, with an interval of **-0.292 to +0.050 percentage points**. The largest-minus-smallest cohort coverage gap falls by only **0.28 percentage points**, with an interval from a **0.74-point reduction to a 0.14-point increase**. Even removing every soft field does not remove the large service gap.

The safe conclusion is descriptive: service differs sharply with starting profile completeness, while these masking interventions do not by themselves prove that the soft fields cause the gap. Arrival order, availability, hard constraints and the simulator's generated population can also contribute.

Evidence: [results/corrected_profile_completeness.json](#) and [results/corrected_profile_completeness_all_masked.json](#)

4. Guarded cardinality and completed outcomes

We first kept the current history score fixed and changed only allocation: greedy versus guarded global allocation. Across 10 fresh seed groups and all six variants, guarded allocation changed MSMI by **-0.0750** with a seed-grouped 95% interval of **-0.2000 to 0.0333**. It shortened median first-service wait by **0.36 days**, with an interval of **0.09 to 0.68 days shorter**, but did not establish an MSMI improvement.

We then removed the uncertain history signal and compared compatibility-greedy allocation with guarded cardinality. Both score **0.2750 MSMI per 100**; the paired difference is **0.0000**, with a seed-grouped 95% interval of **-0.0333 to 0.0333**. Guarded cardinality has slightly lower coverage, **0.32825 versus 0.32867**, and lower mutual acceptances per 100, **5.20 versus 5.30**. It shortens median first-service wait by **0.175 days**, with an interval of **0.042 to 0.325 days shorter**. Under the published ranking order, the tied primary score moves to coverage, so greedy wins this comparison.

The allocator may improve a service measure without improving completed outcomes. We therefore do not use introductions created, internal edge score or runtime as evidence of relationship success.

Evidence: [results/corrected_allocation_outcomes.json](#) and [results/corrected_safe_cardinality_no_history.json](#)

5. Clarification order

The earlier research policy asked for incomplete hard constraints in observable state order. That is reproducible but brittle: a semantically identical JSON request could change who received a question.

We first compared the old rule with a graph-aware challenger that asks about a member with more currently known, potentially feasible partners. It uses only the public state. Equal priorities retain state order.

Across 10 fresh seed groups and six variants, graph-aware order changed MSMI by **-0.0417**, with a seed-grouped 95% interval of **-0.1833 to 0.0833**. It shortened median first-service wait by **1.58 days** (95% interval **0.93 to 2.28**

days shorter) and the 90th percentile by **0.99 days** (95% interval **0.43 to 1.59 days shorter**). Ask cost was unchanged.

The MSMI evidence did not justify the graph-aware score change. The waiting- time improvement remains a service trade-off, not a win on the competition objective.

The submitted mode now uses a narrower correction. It orders eligible asks by arrival day and then opaque member ID. IDs are only a deterministic tie-break; they are not model features. Against the old input-order rule on 20 fresh seed groups and all six variants, the stable rule changed MSMI by **+0.0208 per 100**, with a seed-grouped 95% interval of **-0.0958 to 0.1417**. Coverage changed by **+0.054 percentage points**, with an interval of **-0.100 to +0.221 points**. Ask cost was identical. These score effects are uncertain.

The correctness result is stronger: the submitted action stayed identical across 120 order-only permutations covering members, introductions, feedback, ask logs and constraint arrays. We adopt the stable rule as an order-invariance fix, not as a claimed MSMI improvement.

Evidence: [results/corrected_clarification_order.json](#),
[results/corrected_member_order.json](#) and [results/edge_case_audit.json](#)

6. Edge cases and safeguards

Risk	Safeguard or test
Two variants share one generated world	Bootstrap samples all variants for that seed together.
A seed/variant episode is duplicated or missing	The validator rejects duplicate and incomplete rectangular blocks.
Fit data leaks into calibration or holdout	The three seed sets must be disjoint; validation fails on overlap.
A response window has not matured	All assignments receive 40 follow-up days; any right-censored item is rejected.
No response is mistaken for dislike	The target is explicitly Yes-by-deadline, not latent preference.
A sparse cohort has zero members or zero served members	Rates use safe denominators; wait statistics are omitted when nobody is served.
Missing soft fields create accidental score credit	Missing values add no compatibility points.
A hard constraint is missing or declined	The pair is not declared feasible; clarification never overrides feasibility.
No feasible pairs exist	The policy returns an empty pair list.
Ask budget is below one constraints question	The policy submits no ask.
Member, feedback or constraint-array order changes	The submitted action remains identical; 120 order-only permutations pass.
Scores tie	Canonical pair IDs resolve the tie; IDs never enter the score.
The graph is fully connected	A 200-member dense graph returns 100 non-overlapping pairs in 0.12 seconds in the direct audit.
Feedback is duplicated or has impossible timing	Evidence parsing rejects duplicates and enforces lower and upper time bounds.
Memory comes from an older policy shape	Scalar, list and stale mapping inputs are replaced or safely updated.
Delayed, sparse, shifted, drifting or cold-start conditions	Every study includes all six variants and reports their equal-weight mean.
A complex component improves an internal proxy only	Promotion is based on completed MSMI first, followed by the published tie-break order.
JSON contains invalid numbers	Saved artifacts use strict JSON; the validator rejects NaN and infinity.
Public code accidentally depends on hidden truth	Policy code receives only the documented observable request and runs offline.

The fuller runtime, timing, service and evidence matrix is in [docs/EDGE_CASE_REGISTER.md](#).

An extra matched test changed only equal-score tie ordering so earlier arrivals were preferred. It scored **0.5083** versus **0.5417** for the submitted rule; the difference was **-0.0333**, with a 95% seed-grouped interval from **-0.1167 to +0.0500**. It also made p90 first-service wait **0.4533 days worse**, with an interval from **+0.0833 to +0.9383**. We reject it rather than using a fairness-sounding rule that failed its intended service check.

Across all six corrected algorithm comparisons, the sensitivity audit reports paired seed-level means, exact two-sided sign-flip tests, leave-one-seed-out ranges and approximate 80% detectable effects. For the uncertain history gain, the exact p-value is 0.524 and the approximate detectable effect is 0.0844 MSMI per 100, about four times the observed 0.0208 difference. For the stable member-order correction, leaving out one seed can change the sign. These checks support conservative selection rather than a score claim.

Evidence: [results/corrected_wait_tie.json](#), [results/corrected_sensitivity.json](#)

7. Fixed 33-item closure

The requested closure boundary contains exactly 33 items. Each is marked `fixed`, `measured` or `contract-excluded`; none is `pending`. The only `contract-excluded` item is counterfactual calibration for unassigned pairs. Their outcomes are not observable under the published request, so claiming to estimate them would invent evidence. The policy does not use that quantity.

The executable closure audit checks the row count and states, corrected result files, selected runtime mode, container file whitelist, report-source hash, PDF presence and pinned remote evidence revision. The complete register is [docs/EDGE_CASE_CLOSURE.md](#).

Evidence index

All links below are pinned to the immutable evidence commit used for this report.

Item	GitHub source
Evaluated runtime	policy.py
Container definition	Dockerfile
Independent addendum validator	experiments/validate_corrected_addendum.py
Registered seed families	experiments/seed_registry.json
Reproduction scripts	experiments/
Edge-case closure audit	results/closure_audit.json
Complete result directory	results/
Immutable evidence revision	ccf554f

8. Revised policy claim

The corrected container default is the stable-clarification compatibility- greedy method. We remove both the uncertain history signal and guarded cardinality from the selected runtime. They remain named research ablations so every result can be reproduced. This is a correction to our original selection, not a claim that greedy is universally optimal.

What remains defensible is the boundary of the method:

- 1 clarify only missing reciprocal hard constraints within budget, using a stable arrival-day and opaque-ID order;
- 2 reject every pair that is not reciprocally feasible;
- 3 rank feasible unseen pairs with transparent observed compatibility;
- 4 allocate deterministically;
- 5 use delayed feedback only after it has matured and only if fresh evidence supports the component;
- 6 return valid empty output when no safe action exists.

9. Reproduction and independent checks

```
python experiments/confidence_audit.py --incumbent adaptive_legacy --challenger adaptive --seeds 4001-4020 --
variants all --workers 4 --resamples 20000 --output results/corrected_history_seed_grouped.json
python experiments/component_audit.py --incumbent adaptive_history_greedy --challenger adaptive --seeds 6301-
6310 --variants all --workers 4 --resamples 20000 --output results/corrected_allocation_outcomes.json
python experiments/component_audit.py --incumbent adaptive --challenger adaptive_graph_asks --seeds 6401-6410
--variants all --workers 4 --resamples 20000 --output results/corrected_clarification_order.json
python experiments/component_audit.py --incumbent adaptive_greedy --challenger adaptive_legacy --seeds 6501-
6510 --variants all --workers 2 --resamples 20000 --output results/corrected_safe_cardinality_no_history.json
python experiments/component_audit.py --incumbent adaptive_input_order_greedy --challenger adaptive_greedy --
seeds 7201-7220 --variants all --workers 4 --resamples 10000 --output results/corrected_member_order.json
python experiments/component_audit.py --incumbent adaptive_greedy --challenger adaptive_wait_tie_greedy --
seeds 7401-7410 --variants all --workers 4 --resamples 20000 --output results/corrected_wait_tie.json
python experiments/profile_completeness_audit.py --seeds 6201-6210 --variants all --mask-rate 0.5 --workers 4
--resamples 20000 --output results/corrected_profile_completeness.json
python experiments/profile_completeness_audit.py --seeds 6201-6210 --variants all --mask-rate 1 --workers 4 --
resamples 20000 --input-unmasked results/corrected_profile_completeness.json --output results/
corrected_profile_completeness_all_masked.json
python experiments/directional_calibration.py --training-seeds 6001-6010 --calibration-seeds 6031-6040 --
holdout-seeds 6101-6110 --variants all --workers 4 --resamples 5000 --output results/
corrected_directional_calibration.json
python experiments/edge_case_audit.py --output results/edge_case_audit.json
python experiments/seed_registry.py
python experiments/sensitivity_audit.py --output results/corrected_sensitivity.json
python experiments/validate_corrected_addendum.py
python experiments/closure_audit.py --remote --output results/closure_audit.json
```

The final validator reloads saved JSON, rejects non-finite values, verifies complete seed-by-variant blocks, checks split separation and recomputes every paired mean directly from the raw episode rows. It does not call the analysis helper that produced the results.

Limitations

- All evidence comes from the published synthetic simulator. It may not transfer to real matching outcomes.
- The new component studies contain 10 independent seed groups each. Rare MSMI outcomes still create wide intervals.
- The profile cohorts are observational. Controlled masking removes observable soft information but does not change correlated arrival, availability or hard-constraint patterns.
- The not-ruled-out partner count is an observable opportunity proxy. Missing hard constraints can make it larger than the final feasible partner count.
- Calibration is evaluated on an untouched public-simulator holdout, not on real user probabilities.
- The calibration sample contains assigned introductions under one policy. It does not establish probability quality for every unassigned feasible pair.
- A generated-world interval describes uncertainty across the tested worlds; ten seed groups still leave low power for small effects and rare outcomes.
- Several questions were tested. We show negative results and do not select a method only because one public block looks favourable.

Request for reconsideration

This addendum implements the requested corrections and makes the weaker findings explicit. We respectfully ask that LuminaX be reconsidered using this corrected evidence package. We understand that reconsideration does not guarantee selection.